

Heuristic Process Redesign (Chapter 8.2) – Summary & Key Takeaways

1. Introduction

This section outlines a structured methodology for Business Process Redesign using redesign heuristics – practical rules of thumb derived from successful historical projects. The goal of heuristic process redesign is to generate alternative "to-be" process scenarios from the "as-is" process, in order to achieve targeted performance improvements. These improvements are evaluated using the Devil's Quadrangle, a framework that captures four competing performance dimensions: Time, Cost, Quality, and Flexibility. Because these four dimensions frequently trade off against one another, no single heuristic can improve all of them simultaneously. Applying a heuristic is therefore a deliberate design choice: an analyst must decide which dimension(s) matter most for a given process and accept the corresponding trade-offs elsewhere.

2. The Devil's Quadrangle

The Devil's Quadrangle (Reijers & Limam Mansar) is used to evaluate the impact of a redesign heuristic on process performance along four corners:

- Time – The speed at which cases are handled from start to finish.
- Cost – The monetary resources consumed by executing the process.
- Quality – The value delivered to customers and other stakeholders, in terms of the product/service and the customer's experience of the process.
- Flexibility – The ability of the process to react to variation, exceptions, and changing circumstances.

Because the four corners of the quadrangle pull against each other, redesign is fundamentally about trade-off management: gains on one corner (e.g., faster cycle time) very often come at the cost of another corner (e.g., higher cost or reduced flexibility).

3. Main Stages of Heuristic Process Redesign

- **Initiate** - Understand the as-is process thoroughly, including its activities, actors, resources, and performance issues. This forms the baseline against which redesign options are compared.
- **Design** - Apply relevant redesign heuristics systematically to the as-is process to generate one or more to-be process alternatives. Each heuristic is evaluated for its likely effect on Time, Cost, Quality, and Flexibility before being adopted.

These two stages are highly iterative: candidate redesigns are proposed, assessed against the Devil's Quadrangle, refined or discarded, and the cycle repeats until a satisfactory to-be process emerges.

4. Classification of Redesign Heuristics

Redesign heuristics are grouped according to the seven key process elements they primarily target (customer, business process, business process operation, business process behaviour, organization - structure, organization - population, information, technology, and external environment). These categories are not mutually exclusive - a single heuristic may influence more than one element at once.

This assignment covers the heuristic categories illustrated in the original notes: Customer, Business Process Operation, Business Process Behaviour, Organization, and External / Transformation heuristics.

4.1 Customer Heuristics (Focus: Interaction with customers)

- **Contact Reduction** - Reduce the number of contact points/moments a customer has with the process (e.g. single point of contact, self-service portals, combined billing).
- **Contact Channel** - Redirect customer contact to the most suitable channel - combining or shifting contact from expensive channels (phone, in-person) to cheaper ones (web, email).
- **Case Types** - Separate case types that need to be handled differently, so each type can follow its own optimized sub-process instead of one generic process for all cases.

Effects on the Devil's Quadrangle (Table 8.3):

Heuristic	Time	Cost	Quality	Flexibility
Contact Reduction	+	-	+	•
Contact Channel	+	+	+	-
Case Types	+	+	•	-

Legend: + = positive effect - = negative effect

• = neutral / mixed effect (depends on context)

5. Business Process Operation Heuristics

Focus: the process elements and activities themselves.

- Activity Elimination - Remove non-value-adding or redundant activities from the process to shorten cycle time and cut cost.
- Case-based Work - Combine small, related activities into a single broader activity so they can be executed together.
- Triage - Split a general activity into two or more alternative, simpler activities suited to specific case types (or the reverse: combine similar specific activities into one general activity).
- Activity Composition - Compensate for the removal of a check or control activity with small, low-cost checks embedded elsewhere in the process.

Effects on the Devil's Quadrangle (Table 8.5):

Heuristic	Time	Cost	Quality	Flexibility
Activity Elimination	+	+	-	-
Case-based Work	+	-	+	•
Triage	•	-	+	-
Activity Composition	+	-	•	•

Legend: + = positive effect - = negative effect
 • = neutral/mixed effect (depends on context)

6. Reflection - Exercise 8.5 Association

The heuristics in Table 8.5 for Contextual Redaction on the four dimensions of the Devil's Quadrangle can be summarized as follows: activity elimination consistently reduces time and cost because unnecessary steps are stripped out, but it can also remove implicit quality checks, causing errors and rework to increase. Case-based and triage-style splitting of work improves the fit between an activity and its case type, often raising quality, but the added coordination and specialised routing can raise cost and reduce flexibility. Overall, this exercise demonstrates the central lesson of heuristic redesign: no heuristic is universally "good" - its value depends entirely on which corner of the Devil's Quadrangle the organisation is prioritising for that process.

7. Business Process Behaviour Heuristics

Focus: control-flow / logic of the process.

- Resequencing - Move activities to a better point in the process - e.g., perform tasks in parallel rather than in sequence, or move a check earlier so exceptions surface sooner.
- Parallelism - Execute independent activities in parallel instead of sequentially, so the overall case-handling time is reduced.
- Knock-out - Order the checks within a process by increasing effort and decreasing probability of termination, so cheap high-yield checks run first.
- Exception - Remove rarely-used exception-handling paths from the normal process flow and treat them separately, keeping the main flow lean.

Effects on the Devil's Quadrangle (Table 8.3, contd.):

Heuristic	Time	Cost	Quality	Flexibility
Resequencing	+	+	•	•
Parallelism	+	-	•	-
Knock-out	-	+	•	-
Exception	+	-	+	-

Legend: + = positive effect - = negative effect
 • = neutral / mixed effect (depends on context)

8. Organization Heuristics

Focus: who performs the work and how teams/roles are structured.

- Case Assignment / Case Manager - Assign work to smaller, dedicated, cross-functional teams rather than large functional departments, so a case is handled by fewer people.
- Flexible Assignment - Make work more flexible by cross-training staff and organizing generalist teams rather than narrow specialists, so work can move where capacity exists.
- Case Manager - Give a single case manager end-to-end responsibility and a single point of contact for a case, improving accountability and customer experience.